

React.js Complete Course

LearnSphere Official Notes

Chapter 1: Introduction to React

React is a declarative, efficient, and flexible JavaScript library for building user interfaces. It lets you compose complex UIs from small and isolated pieces of code called "components".

Why use React?

- Component-based architecture
- Virtual DOM for performance
- Declarative syntax
- Strong community and ecosystem

Chapter 2: Components & Props

Components are independent and reusable bits of code. They serve the same purpose as JavaScript functions, but work in isolation and return HTML.

Example of a Functional Component:

```
function Welcome(props) {  
  return <h1>Hello, {props.name}</h1>;  
}  
  
function App() {  
  return (  
    <div>  
      <Welcome name="Sara" />  
      <Welcome name="Cahal" />  
      <Welcome name="Edite" />  
    </div>  
  );  
}
```

Chapter 3: State & Hooks

Hooks represent a fundamental shift in how we write React components. They let you use state and other React features without writing a class.

The `useState` Hook:

```
import React, { useState } from 'react';

function Counter() {
  // Declare a new state variable, which we'll call "count"
  const [count, setCount] = useState(0);

  return (
    <div>
      <p>You clicked {count} times</p>
      <button onClick={() => setCount(count + 1)}>
        Click me
      </button>
    </div>
  );
}
```

Happy Coding with React!